Malaysia: an emerging ESS market—grid-side tenders break new ground, while demand from C&I and data centers unlocks further growth

Load growth and energy transition: dual foundations underpinning Malaysia's energy storage development

As a key economy in ASEAN and a regional hub for manufacturing, shipping, and digital infrastructure, Malaysia has seen steadily rising electricity demand in recent years. According to Tenaga Nasional Berhad (TNB), Malaysia's national electricity utility company, electricity demand in Peninsular Malaysia reached a record high of 131 TWh in 2024 and is expected to increase further to 135 TWh in 2025, with an average annual growth rate of about 4.9% during 2025–2027 (accounting for over 75% of total national demand).

The combined effects of industrial upgrading, expanding commercial loads, and the rapid deployment of energy-intensive applications such as data centers are not only driving overall electricity consumption upward, but also generating structural demand for grid peak-shaving, reserve capacity, and localized grid support. Meanwhile, Malaysia's three major power systems—Peninsular Malaysia, Sabah, and Sarawak—operate as largely independent grids with limited interconnection capacity. This structural constraint restricts cross-regional power balancing and prevents system-wide resource optimization to mitigate load fluctuations. Together, these factors form a solid foundation underlying the large-scale development of the energy storage market in Malaysia.

From a supply mix perspective, Malaysia's power system remains heavily dominated by fossil fuels, while a clear long-term pathway for renewable energy expansion has been established. According to TNB, renewable energy accounted for only 23% of its domestic installed capacity in 2024, with fossil fuels comprising close to 80%. In terms of actual generation, however, renewables contributed only around 9% of total output, revealing a significant gap between installed capacity share and actual generation share.

This disparity suggests that, while renewable energy has gained a foothold at the capacity level, its effective contribution to electricity supply remains limited. System operations continue to rely predominantly on dispatchable sources such as coal and natural gas.

Against this backdrop, the Malaysian government's target of increasing the share of renewable energy in the power mix to 70% by 2050 further elevates the strategic importance of energy storage in the low-carbon transition of the power

system. Energy storage mitigates intermittency from renewable sources such as solar PV, alleviates integration and curtailment challenges, and ensures the smooth grid connection of large-scale renewable capacity.

At the same time, as conventional thermal power plants are gradually phased out, energy storage will become a core provider of essential grid services, including frequency regulation, peak shaving, and contingency reserves. Furthermore, given Malaysia's structurally fragmented power system, energy storage is indispensable for enabling coordinated optimization across generation, grid, load, and storage.

Collectively, these factors position energy storage as a central pillar in Malaysia's power system transformation, unlocking broad-based, multi-scenario growth potential over the medium to long term.

Policy framework gradually taking shape, with utility-scale projects leading deployment

At the institutional level, Malaysia has established a policy framework for energy storage development that spans mid- to long-term transition, distributed applications, and corporate green power use cases.

At the strategic level, the National Energy Transition Roadmap (NETR) sets a target of achieving a 70% renewable energy share in installed capacity by 2050, thereby defining long-term growth opportunities for energy storage in grid peak-shaving, reserve capacity, and system flexibility.

In the distributed segment, the Net Energy Metering (NEM) 3.0 allocates a total quota of 2,500 MW for 2021–2025, covering residential, government, and commercial & industrial (C&I) users. This scheme continues to drive the expansion of rooftop solar, while creating favorable conditions for the co-deployment of energy storage.

On the corporate side, the Corporate Renewable Energy Supply Scheme (CRESS) has incorporated energy storage into its implementation framework, formally enabling storage applications in corporate green power consumption and reliable power supply scenarios.

Overall, Malaysia has largely established the institutional foundation for energy storage development. However, a fully developed and standalone mechanism for capacity remuneration and ancillary services pricing for energy storage has yet to be put in place.

Key policy framework for energy storage in Malaysia

Policy category	Policy name	Core content	Implications for energy storage
Strategic target	NETR	Targets a 70% share of renewable energy in installed capacity by 2050.	Provides a basis for medium- to long-term system flexibility needs, helping define the direction for energy storage capacity expansion.
Behind-the-meter policy	NEM 3.0	Total quota of 2,500 MW for 2021–2025, including NEM Rakyat (700 MW), NEM GoMEn (100 MW), and NOVA (1,700 MW)	Accelerates distributed PV adoption, supporting the development of C&I storage, self-consumption models, and peak-shaving applications.
Corporate green power scheme	CRESS	Allows companies to procure and wheel renewable electricity through the public grid, while preserving room for supporting energy storage applications to stabilize green power supply.	Creates a institutional pathway for "green power trading + storage," supporting energy storage deployment in the C&I and industrial park segments

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At the project level, the rollout of the MyBeST program marks a critical breakthrough for Malaysia's utility-scale energy storage market, signaling a transition from policy planning to large-scale deployment. On December 19, 2025, the Energy Commission of Malaysia, also known as the Suruhanjaya Tenaga (ST), announced the shortlisted bidders for the project, confirming the development of four grid-connected battery energy storage system (BESS) projects in Peninsular Malaysia, with a total capacity of 400 MW / 1,600 MWh. The projects are scheduled to be commissioned in 2027.

As the largest grid-scale energy storage project in Malaysia to date—awarded through an international competitive tender—MyBeST aims to reduce the grid's reliance on conventional fossil-fuel-based peaking plants, lower overall system operating costs, and support the long-term renewable energy targets outlined in the NETR.

The project establishes, for the first time within the Peninsular Malaysia grid system, standardized frameworks for utility-scale energy storage, including tendering procedures, grid connection technical requirements, and dispatch and operational rules. It also removes key institutional barriers for subsequent large-scale deployment, positioning MyBeST as a benchmark project for the commercialization and scaling of Malaysia's energy storage market.

Key Energy Storage Projects Implemented in Malaysia

Project	Location	Power output	Capacity	Project status	Date of announcement /commissioning
MyBeST	Peninsular Malaysia	400 MW	1,600 MWh	Shortlisted, expected to be commissioned in 2027	2025.12
Sejingkat BESS	Kuching, Sarawak	60 MW	82 MWh	Commissioned	2025.02
Lahad Datu BESS	Lahad Datu, Sabah	100 MW	400 MWh	Commissioned	2024.09 / 2025.12

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Expansion of data center clusters is expected to drive additional demand for energy storage

Malaysia's data center sector is entering a phase of rapid expansion, positioning it as one of Southeast Asia's most dynamic internet data center (IDC) markets and a key driver of power load growth and energy storage demand. According to the Malaysian Investment Development Authority (MIDA), Malaysia had 54 operational data centers as of the end of 2024, with total IT load reaching 504.8 MW. Newly contracted IT load in 2024 alone reached 429 MW, nearly matching the cumulative operational capacity added in prior years.

Supported by its proximity to Singapore, relatively low power costs, and favorable foreign investment policies, Malaysia has emerged as a key hub for global players expanding digital infrastructure in Southeast Asia.

Johor has emerged as the primary growth hub, with Sedenak Tech Park, Johor Bahru, and Iskandar Puteri forming large-scale IDC clusters. The presence of leading players such as Yondr, PDG, and AirTrunk is accelerating the shift toward more clustered, park-based development.

Key Data Center Projects in Malaysia

Located city	Project name	Scale	Status	Implications for energy storage
Kulai / Sedenak Tech Park	Yondr Johor Campus	Total planned IT load of 300 MW	Under development, phased construction	High-density load in tech parks, supporting demand for on-site storage, backup power, and localized grid support
Kulai / Sedenak Tech Park	PDG JH1 Campus	Total capacity of 150 MW	Phase 1 delivered / commissioned	AI-ready campus project, well-suited for storage deployment in backup power, power quality management, and peak shaving
Johor Bahru	AirTrunk JHB1	150 MW	Commissioned	Continuous high-load operation, driving practical demand for on-site storage and high-reliability power configurations
Iskandar Puteri	AirTrunk JHB2	>270 MW	Under construction / progressing	Concentrated large-scale load growth, benefiting grid-side storage, on-site storage, and renewable-plus-storage solutions

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The rapid expansion of data centers is emerging as a key driver accelerating Malaysia's energy storage market from policy-led development toward demand-driven

commercialization. The associated demand for energy storage extends beyond renewable integration, evolving into rigid, multi-scenario requirements across the entire value chain.

According to TNB, as of August 2025, Peninsular Malaysia had secured 47 planned data center projects, with a total planned IT load of 6.7 GW. Incremental power demand from these projects is expected to reach nearly 13 GW by 2030 and exceed 20 GW by 2040, positioning data centers as the primary driver of electricity demand growth in Malaysia over the next decade.

These large-scale, Tier-1 loads are characterized by high density, continuous operation, and extremely stringent requirements for power reliability. Coupled with Malaysia's structurally fragmented grid system—where Peninsular Malaysia, Sabah, and Sarawak operate as largely independent networks with limited interconnection—this further amplifies the structural demand for energy storage deployment.

In practice, data center-driven demand for energy storage spans three key application areas:

- Behind the meter: providing backup power to ensure uninterrupted operations and enhance supply reliability;
- Grid side: enabling peak shaving, supporting energy arbitrage, alleviating transmission constraints, and enhancing grid resilience;
- Renewable integration: facilitating the integration of distributed PV and supporting green power consumption, helping corporates meet decarbonization targets.

Conclusion and outlook

Malaysia's electrochemical energy storage market has moved beyond the initial stage characterized by scattered pilot projects without clear deployment pathways, and has entered an early growth phase driven by four key factors: sustained electricity demand growth, the low-carbon energy transition, progressive policy framework development, and the emergence of scalable application scenarios.

In the near term, the clearest opportunities for large-scale deployment will continue to stem from grid-side, utility-scale projects, as exemplified by MyBeST. Over the medium term, with continued expansion of PV capacity under NEM 3.0 and the opening of corporate renewable power procurement under CRESS, supporting market mechanisms are expected to gradually mature. This will likely translate into more stable and

sustainable demand for energy storage across C&I and industrial park applications.

Looking ahead, data centers are likely to become the most significant driver of growth in Malaysia's energy storage demand over the next two to five years. As major data center clusters—particularly in Johor—begin to treat power reliability, peak load management, and renewable energy integration as baseline requirements for project deployment, Malaysia's energy storage market is expected to evolve from a grid-led development model toward a phase of multi-scenario coordinated expansion.